

FORCES – FORCES & MOTION

KEY VOCABULARY

Resultant force: the single force that has the same effect as all forces.

Weight: the force of gravity on a mass; $W = m \times g$.

Acceleration: the rate of change of velocity, in m/s^2 .

Terminal velocity: steady speed when drag balances weight.

COMMON MISCONCEPTIONS

~~— A moving object always has a force pushing it.~~

✓ At constant speed, forces are balanced.

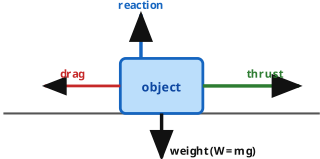
~~— Mass and weight are the same thing.~~

✓ Mass is in kg; weight is a force in newtons.

~~— Heavier objects always fall faster.~~

✓ Without air, all objects fall at the same rate.

FORCES ON A MOVING OBJECT



If thrust = drag and reaction = weight, the resultant force is **zero**.

1 WEIGHT

A 6 kg mass sits on Earth ($g = 9.8 \text{ N/kg}$). Calculate its weight. [2 marks]

Answer: _____ N

2 NEWTON'S SECOND LAW

A resultant force of 12 N acts on a 4 kg trolley. Calculate its acceleration ($F = m a$). [2 marks]

Answer: _____ m/s^2

3 RESULTANT FORCE

Explain what happens to a car's motion when the driving force equals the drag. [2 marks]

4 TERMINAL VELOCITY

Explain why a skydiver reaches a terminal velocity. [3 marks]

5 BALANCED FORCES

Complete the sentence using the word bank. [2 marks]

When forces are balanced the resultant force is _____ and the object moves at a _____ velocity.

zero

constant

increasing

non-zero

6 STOPPING DISTANCE

Stopping distance = thinking distance + braking distance. State **one factor** that increases each. [2 marks]
